Multimodal Transformer Network for Hyperspectral and LiDAR Classification

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***Abstract*— The land cover classification of single-modal remote sensing (RS) data has recently reached a bottleneck. The joint use of multimodal RS data to improve the classification performances has received much attention. Convolutional neural networks are powerful tools in feature extraction and contextual modeling. While they have attendant drawbacks to capture the sequence attributes of spectral signatures and struggle to acquire discriminative spectral–spatial features from a global perspective due to limitations inherent in their network backbones. The transformer backbone is a promising approach for addressing these challenges and generating novel insights in the multimodal RS image classification. In this article, we present a new model called multimodal transformer network (MTNet) that leverages transformer advantages to capture both the specific and shared characteristics of hyperspectral (HS) and light detection and ranging (LiDAR) data. HS images contain a wide range of bands with rich spectral information and LiDAR data provide accurate elevation information without affecting by environmental factors. The well-designed module HS spectral transformer can learn spectrally local sequence information from neighboring bands of HS images, yielding groupwise spectral embeddings comprising rich diagnostic information about land covers. Furthermore, the HS and LiDAR spatial transformers aim to mine the pixelwise feature embedding relationships in a global manner, capturing spatial and elevation information of HS and LiDAR, respectively. Finally, the feature embedding tokens of two modalities are integrated jointly and a new transformer encoder is redesigned to explore the shared spatial characteristics between the two modalities. We evaluate the classification performances of the proposed MTNet on three public HS-LiDAR datasets by conducting extensive experiments, exhibiting superiority over conventional classifiers and state-of-the-art networks.**

***Index Terms*— Class tokens, hyperspectral (HS) images, light detection and ranging (LiDAR), multimodal image classification, transformers.**

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NOMENCLATURE

RS	Remote sensing.
HS	Hyperspectral.
LiDAR	Light detection and ranging.
CNN	Convolutional neural network.
MLP	Multilayer perceptron.
MTNet	Multimodal transformer network.
NLP	Natural language processing.
RNN	Recurrent neural network.
SA	Self-attention.
ViT	Vision transformer.
MHA	Multihead attention.
HSeF	HS spectral transformer.
HSaF	HS spatial transformer.
LSaF	LiDAR spatial transformer.
HLCF	HS and LiDAR collaboration transformer.
SVM	Support vector machine.
RF	Random forest.
MLR	Multinomial logistic Regression.
KNN	K-Nearest Neighbor.
ROI	Region of interest.

I. INTRODUCTION

WITH the recent rapid development of RS technology, researchers have access to a significant amount of Earth observation data for analysis and processing, which plays a crucial role in various fields, including agricultural production [1], mineral exploration [2], and urban planning [3]. As one of the hotspots in RS, HS images have gained increasing attention due to their rich diagnostic spectral bands of land covers [4]. Numerous studies based on HS images have been carried out, including image classification [5], [6], super-resolution [7], [8], target detection [9], [10], [11], and change detection [12], [13], [14], [15]. However, with the further development of RS technology, HS data encounter bottlenecks when applied to complex scenes. For instance, HS images are sensitive to external environmental factors such as atmosphere and light, making them susceptible to interference that can reduce the reliability of diagnostic spectral information [16]. Consequently, it has been found that using single-source RS data is often insufficient for geographic analysis due to significant interference factors [17], [18]. Therefore, multisource RS data have been considered to enhance performances.

Coincidentally, LiDAR is naturally complementary to HS data applications [19]. LiDAR, as an active RS technology,

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is distinct from RS based on spectral imagery as it provides concrete elevation information about the land covers. In addition, LiDAR is unaffected by environmental factors, which allows for accurate observations of land covers at all times. Furthermore, LiDAR imagery will not be interfered with by the cloud cover to contain shadow regions.

Significant progress has been made in research into the classification of multisource data [20], [21], [22], [23]. Many traditional methods, such as morphological extended attribute profiles and kernel learning methods, have been proposed to focus on feature extraction. Liao et al. [24] proposed a generalized graph-based fusion method that integrated the HS spectral features and the dimensionality-reduced morphological profile features from HS and LiDAR data. Jia et al. [25] proposed a multiple feature-based superpixel-level decision fusion method for HS and LiDAR data classification. The method utilized various Gabor filters to extract features from HS and LiDAR data. Subsequently, a weighted majority voting-based decision fusion strategy was incorporated to enhance the joint utilization of the multisource data. However, the aforementioned early works primarily depend on artificially designed features that incorporate subjective ideas and cannot adaptively capture the intrinsic features of multimodal data.

Recently, CNNs have shown great potential in multimodal data classification due to their powerful feature representation ability [26], [27], [28]. Different from conventional methods, CNN-based models do not require handcrafted features based on expert knowledge. Instead, they perform adaptive feature extraction through end-to-end networks. Many recent attempts have been made using CNN-based methods for multimodal data classification. A preliminary attempt was made from the perspective of spectral band correlation. To be specific, Morchhale et al. [29] regarded the LiDAR-based DSM as the additional band of HS and sent them into the CNN model to learn mutual features. Likewise, Wu et al. [30] designed a cross-channel reconstruction module named CCRNet, which learned compact fusion representation of HS and LiDAR data through reconstruction strategy across modalities more effectively. Hong et al. [31] adopted a novel semisupervised cross-modal network to learn features from limited samples in multimodal RS applications. Zhao et al. [32] proposed an effective hierarchical random walk network to learn boundary localization and spatial consistency between HS and LiDAR data. In [33], a simple and effective attention fusion model based on disentangled nonlocal network was proposed for joint classification tasks.

The previous models considered heterogeneous data as a unified entity and extracted mutual features. Moreover, various studies have proposed feature extraction modules for each heterogeneous data type and integrated the particular features. Roy et al. [34] introduced a joint feature learning mechanism for HS and LiDAR classification with CNN and spatial morphological blocks. Specifically, the 3-D CNN was applied for extracting discriminative spectral–spatial feature representation. Simultaneously, the spatial morphology was designed to capture the information on the height and shapes of land covers for the compensation of CNN models. The obtained features were fused using an additive operation before the final classification. Chen et al. [35] utilized two CNN streams to extract features from HS and LiDAR data individually. Then, they designed a DNN with fully connected layers to integrate the heterogeneous features obtained by the previous CNNs. Finally, a logistic regression classifier was employed to generate the ultimate classification map. Meanwhile, Hang et al. [36] proposed an efficient two-stream coupled CNN structure to fuse HS and LiDAR data. One stream aimed to capture spatial–spectral features of the HS data, and the other focused on obtaining elevation information of the LiDAR data. The feature- and decision-level fusion methods were adopted simultaneously in the fusion process of the two heterogeneous data. Zhang et al. [37] proposed a classification method that focuses on the process of structural information transmission and physical properties alignment between heterogeneous data called structural optimization transmission.

Due to the significant disparity between HS and LiDAR data, numerous methods concentrate on devising efficient feature extraction operators and suitable strategies for information fusion-based classification. Hong et al. [38] introduced a general multimodal deep learning framework that deals with complex scenes requiring precise classification. They also investigated different fusion strategies and extractors for spatial–spectral and pixel-based classification. Ghamisi et al. [39] utilized extinction profiles to extract spatial and elevation information from HS and LiDAR, fused the features, and fed them into CNNs with logistic regression to generate the final classification maps. Wang et al. [40] proposed the fusion network AM^{3}Net, aiming to address many methods that ignore the complementarity of the HS images' spectral channels and LiDAR spatial information. The approach utilizes the involution operator to explore the relationships between channels, followed by designing a spectral–spatial mutual-guided module for collaborative transmission between spectral features and spatial information. Due to the significant domain differences among multiple heterogeneous sources, attention mechanisms have been introduced to improve multimodal classification. Mohla et al. [41] used SA to generate an attention map for the HS modality and then utilized a cross-attention module to exploit the LiDAR-derived attention map, which highlights the spatial characteristics of HS named FusAtNet. In [42], a multiattentive hierarchical fusion network was proposed, which utilized modality attention and SA modules to generate integrated modality features and specific features. The obtained two features were fused to construct the classification framework.

Despite the excellent performance of CNNs, they have limitations in dealing with long sequence attributes of spectral signatures due to their inherent network backbone. Due to the powerful representation ability of ViTs, Hong et al. [43] developed a spectral transformer to extract spectral discriminative features from groupwise bands. Qing et al. [44] proposed an end-to-end transformer, which used spectral and SA mechanisms to extract spectral-spatial features. Roy et al. [45] introduced a novel morphological transformer for HS image classification, employing a learnable spectral and spatial morphological network that works with the attention mechanism

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to integrate structural and shape information from both the HS token and class token. Xiu et al. [46] developed a multisource attention network with discriminative graphs and informative entitles for HS and LiDAR classification. The semisupervised graph transductive module was designed to construct a multimodal adjacency matrix explicitly to underline the relevance among pixels. Moreover, the self-supervised feature extraction module was proposed to alleviate the overfitting and oversmoothing problems. Xue et al. [47] proposed a hierarchical ViT architecture for fusing HS and LiDAR data. The model utilized the powerful modeling capability of long-range dependencies and the strong generalization ability to capture discriminative features across modalities. Likewise, a new transformer backbone was proposed to extract feature representations from multisource data, and class tokens were used for final classification [48].

Based on the above analysis, we are meant to develop a novel transformer-based network for HS-LiDAR joint classification. Our primary motivation is to leverage transformers' superior global feature representation capability and focus on extracting the specific and shared features from HS and LiDAR data. Unlike CNNs, transformers employ SA techniques, making them an innovative backbone network that can efficiently process and analyze sequential data. It is particular because the characteristic transformers can be capable of learning spectral sequences from HS data. In addition, we revisit the transformer structures from the spatial features extractor perspective, enabling them to focus on position relationships between spatial pixels from HS and LiDAR data in a global manner. The main contributions of this article are summarized as follows.

1) We propose a novel end-to-end framework, called the MTNet, for accurate joint classification of HS and LiDAR data. This framework is a significant attempt to explore the effectiveness of transformer structures for multimodal RS data classification.
2) We devise the HSeF for extracting spectral characteristics unique to HS, which is capable of learning spectrally local sequence information from neighboring bands of HS images, yielding groupwise spectral embeddings that contain rich diagnostic information about land covers.
3) Despite the effectiveness of the HSeF in capturing the unique HS attributes along the spectral sequences, we also design similar spatial transformer encoders to model the pixelwise embeddings, which allows us to globally capture the relationships among pixelwise feature embeddings and effectively obtain the spatial and elevation information of both HS and LiDAR data. In addition, we integrate token embeddings from both heterogeneous data sources into a new encoder to extract shared intermodality features and reduce the impact of the semantic gap.
4) We conducted experiments to evaluate the classification performances of our proposed model on three popular datasets (i.e., Houston, Trento, and MUUFL). We also performed extensive ablation and parameter-sensitive experiments to analyze the contribution of each component and select the optimal parameters. The results demonstrate the effectiveness and superiority of our proposed approach both qualitatively and quantitatively.

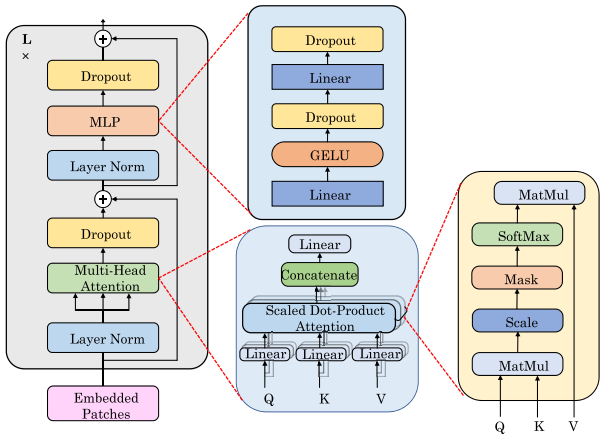


Fig. 1. Schematic of the transformer encoder structure. MLP module consists of five layers: dropout, linear, dropout, Gaussian error linear unit (GeLU), and linear. The MHA module comprises multiple SA heads to enhance the feature representation ability.

The rest of this article is organized as follows. Section II gives preliminary knowledge about the transformers, and Section III details the proposed MTNet. Extensive experiments and results discussions are conducted in Section IV. Section V gives broad conclusions and future research directions.

II. Preliminary of Transformers

A. Brief Review of Transformer

Transformers [49] have emerged as prominent tools in processing sequence-to-sequence problems in NLP [50], such as text classification [51], sentiment analysis [52], machine translation [53], and information retrieval [54]. Unlike traditional RNNs [55], transformers do not rely on sequential dependence. Instead, they employ the novel SA mechanism to capture global information, including long-term dependencies, for units at any position. The network structure has advanced the development of time-series data processing models and gained attention in image processing. Recent breakthroughs, such as the ViT [56], have exhibited exceptional and even surpassed the state-of-the-art performance of CNNs in various computer vision (CV) tasks [57], [58], [59]. Consequently, ViT has provided new inspiration for vision-related tasks.

B. SA and Multihead Attention

SA is a critical mechanism in transformers that enables them to capture global dependencies between different sequence parts. In contrast to RNNs that process sequences in a fixed order, transformers use SA to compute a weighted sum of all the input embeddings at each position, where the weights are learned dynamically based on the similarity between each input and all other inputs in the sequence.

Fig. 1 shows the structure of the transformer encoder, which relies on the SA mechanism as the core component of the architecture. Specifically, the SA can be performed in the following steps.

1) Given the input sequence data x with length of m, where $x_i, i = 1, \ldots, m$.

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TABLE I
DIFFERENT NOTATIONS USED IN THE PROPOSED MODEL

Notation	Definition	Type	Size
n	the number of neighbor pixels	Scalar	1×1
m	the number of all pixels in HS images	Scalar	1×1
d	the number of pixels in an image patch	Scalar	1×1
c	the number of spectral bands	Scalar	1×1
$\boldsymbol{x}$	spectral signature of a HS pixel	Sector	$1\times c$
x_i	reflectance value in the i-th band location	Scalar	1×1
$\boldsymbol{X}$	spectral signatures of d pixels	Matrix	$d\times c$
$g(x)$	overlapping grouping operation	-	-
$\lfloor\cdot\rfloor$	the round operation	-	-
$X^{\dagger}$	the HS patch cubes	Matrix	$c\times h\times w$
$\boldsymbol{P_1}$	random matrix for poistion embedding for HSeF	Matrix	$d\times(c+1)$
$\boldsymbol{P_2}$	random matrix for poistion embedding for HSaF	Matrix	$c\times(d+1)$
$\boldsymbol{P_3}$	random matrix for poistion embedding for LSaF	Matrix	$c\times(d+1)$
$\boldsymbol{P_4}$	random matrix for poistion embedding for HLCF	Matrix	$c\times(2d+1)$
$\hat{\boldsymbol{X}}$	the unfolded patches along with the spatial dimension	Matrix	$c\times d$
$\dot{\boldsymbol{X}}$	the unfolded patches and rearrange	Matrix	$d\times c$
$\widehat{\boldsymbol{x_i}}$	embeddings in the i-th band location for HSeF	Vector	$d\times1$
$\widehat{\boldsymbol{x_i}}$	embeddings for the i-th pixels for HSaF	Vector	$c\times1$
$\widetilde{\boldsymbol{x_i}}$	embeddings for the i-th pixels for LSaF	Vector	$c\times1$
$\hat{\boldsymbol{x_i}}$	embeddings for the i-th pixels for HLCF	Vector	$c\times1$
$\boldsymbol{F_l}$	the feature representations for the l-th layers for HSeF	Matrix	$(c+1)\times d$
$\boldsymbol{G_l}$	the feature representations for the l-th layers for HSaF	Matrix	$(d+1)\times c$
$\boldsymbol{H_l}$	the feature representations for the l-th layers for LSaF	Matrix	$(d+1)\times c$
$\boldsymbol{Z_l}$	the feature representations for the l-th layers for HLCF	Matrix	$(2d+1)\times c$

2) Obtain the feature embedding $\boldsymbol{\alpha_i}$ on each vector $\boldsymbol{x_i}$ using a shared matrix $\boldsymbol{W}$.
3) Initialize three different matrices $\boldsymbol{w_q}$, $\boldsymbol{w_k}$, and $\boldsymbol{w_v}$ randomly, which are then multiplied with each embedding to yield three vectors: query ($Q=[\boldsymbol{q}_1,\ldots,\boldsymbol{q}_m]$), key ($K=[\boldsymbol{k}_1,\ldots,\boldsymbol{k}_m]$), and value ($V=[\boldsymbol{v}_1,\ldots,\boldsymbol{v}_m]$).
4) Compute the attention score s between each Q and K vector using the inner product and adopt the scaled score to stabilize gradients by normalization, i.e., $s_{i,j}=\boldsymbol{q_i}\cdot\boldsymbol{k_j}/(d)^{1/2}$, where d is the dimension of $\boldsymbol{q}_i$ or $\boldsymbol{k}_i$.
5) Perform the softmax activation function on attention scores to map the range of each element value between (0, 1).
6) Generate the attention representations $z=[z_1,\ldots,z_m]$ by multiplying the attention score $s_{i,j}$ and value V.

Therefore, the overall calculation process of SA can be formulated as follows:

$$z=\text{Attention}(Q,K,V)=\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \tag{1}$$

More specifically, the transformers enhance the capacity to capture complex dependencies and features from inputs by employing the MHA mechanism, which consists of stacking multiple SA layers to obtain attention representations and concatenate them into a larger feature matrix. The formula for the MHA can be expressed as follows:

$$\text{MHA}(Q,K,V)=\text{Concat}(\text{head}_1,\ldots,\text{head}_n)\boldsymbol{W^O} \tag{2}$$

where $\boldsymbol{W^O}$ is used to ensure the feature dimension of the output identical to that of the input data and n refers to the number of attention heads of the MHA. By computing attention in parallel across multiple heads, the transformers can capture more complex relationships between different positions in the input sequence. In addition, MHA allows the model to learn different patterns in the data across different subspaces, which can lead to improved performances on a wide range of downstream tasks in NLP or CV.

III. METHODOLOGY

A. Overview of the Proposed MTNet

We propose a novel ViT-based baseline network to extract distinctive spectral characteristics for HS data and spatial-elevation information for HS and LiDAR data to facilitate fusion and accurate land cover classification of two heterogeneous data sources. To be specific, we designed four different transformer structures. Considering the rich diagnostic information of land covers in spectral sequence signatures, we developed the HSeF to learn spectrally local sequence information from neighboring bands of HS images, yielding groupwise spectral embeddings that contain rich diagnostic information about land covers. In addition, two modal-specific transformer structures, HSaF and LSaF, were devised to mine the unique spatial and elevation features of HS and LiDAR data. Furthermore, the collaborative transformer structure was designed to mine the shared spatial features between the different modalities, i.e., HS and LiDAR. Overall, we considered both modalities' distinctive and common characteristics and designed corresponding transformer structures to capture their shared and individual features. The structure of the proposed MTNet is shown in Fig. 2, while Table I details the notations used in the proposed MTNet.

B. HS Spectral Transformer

HS images are abundant in spectral channels, densely sampled from the spectrum, and capable of producing continuous spectral signatures that provide rich information for land cover identification. To accurately classify the materials, it is

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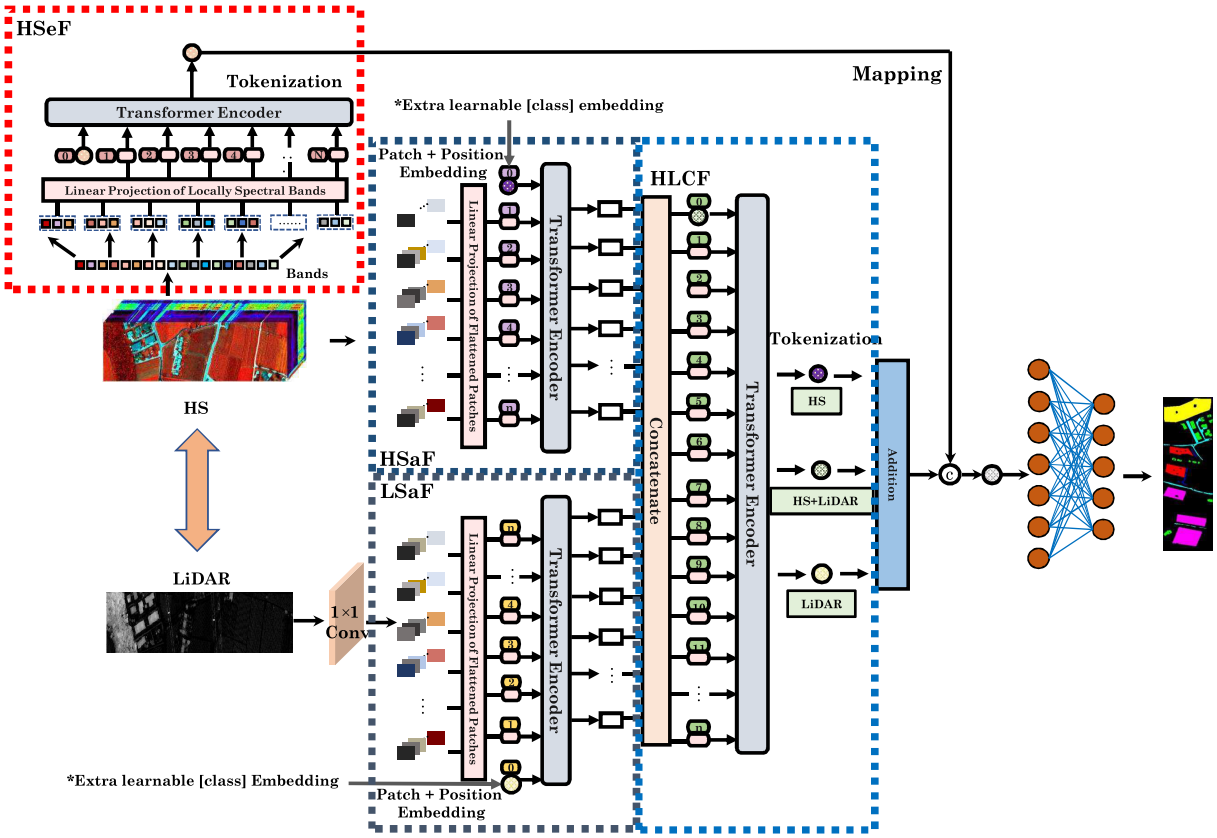


Fig. 2. Schematic of the proposed MTNet for the joint classification of HS and LiDAR data. The MTNet consists of multiple transformer-based modules, i.e., HSeF, HSaF, LSaF, and HLCF.

crucial to capture the locally detailed absorption or changes of such spectral signatures. Inspired by SpectralFormer [43], we propose a new approach to learning groupwise spectral embeddings instead of bandwise representations in HS images. Given a spectral sequence (consisting of one pixel) $\boldsymbol{x} = [\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_c}] \in R^{1\times c}$, where c is the number of spectral channels, we need to select neighbor overlapping bands to learn the feature embedding from locally spectral profiles. The formula can be denoted as

$$\boldsymbol{X} = g(x) = \left[\boldsymbol{x_1}, \ldots \boldsymbol{x_q}, \ldots, \boldsymbol{x_c}\right] \in \mathbb{R}^{m\times c} \tag{3}$$

where $\boldsymbol{x_q} = [\boldsymbol{x_{q-\lfloor(n/2)\rfloor}}, \ldots, \boldsymbol{x_q}, \ldots, \boldsymbol{x_{q+\lfloor(n/2)\rfloor}}]^T \in R^{n\times 1}$, $\lfloor\cdot\rfloor$ denotes the round operation, n and m are the number of neighboring pixels and all pixels of HS images, respectively, n is empirically set to 3, and $g(\cdot)$ denotes the overlapping grouping operation regarding $\boldsymbol{x}$.

The input HS data are given as patch cubes $X^{\dagger} \in R^{c\times h\times w}$, where h and w are the height and length of the patch, respectively. We can unfold these patches along the spatial dimension

$$\hat{\boldsymbol{X}} = \left[\vec{\boldsymbol{x_1}}, \ldots, \vec{\boldsymbol{x_i}}, \ldots, \vec{\boldsymbol{x_c}}\right] \tag{4}$$

where $\vec{\boldsymbol{x_1}} \in R^{d\times 1}$ $(d = h \times w)$ denotes the unfold patch for the ith band. This input approach effectively preserves spatial information during network learning while also considering groupwise spectrally contextual information. The spectral feature embeddings are then sent into the transformer encoder. First, we have to concatenate an extra learnable class embedding, which can be formulated by

$$\boldsymbol{F_1} = \text{Concat}\left(\boldsymbol{C_1}, \hat{\boldsymbol{X}}\right) = \left[\boldsymbol{x}_{\text{hsef_cls}}, \vec{\boldsymbol{x_1}}, \ldots, \vec{\boldsymbol{x_i}}, \ldots, \vec{\boldsymbol{x_c}}\right]. \tag{5}$$

Then, we utilize positional embeddings for retaining the sequential information of spectral embedding tokens and encoding the position of each token. This approach is similar to the transformer structures used in NLP. By incorporating positional embeddings, our model can leverage the advantages of MHA and focus on different groupwise spectral embeddings along the spectrum dimension. To achieve this, we add the positional embeddings, denoted as $\boldsymbol{P_1}$, into the obtained spectral embedding tokens, denoted as $\boldsymbol{F_1}$

$$\boldsymbol{F_2} = \boldsymbol{F_1} + \boldsymbol{P_1}. \tag{6}$$

Specifically, we denote the positional embeddings as $\boldsymbol{P_1} = [\boldsymbol{p_1}, \ldots, \boldsymbol{p_{c+1}}]$, which is a $d \times (c+1)$ random matrix. This matrix is used to mark the position of each token during model training. The feature embedding $\boldsymbol{F_2}$ is then fed into a transformer encoder composed of alternating MHA and MLP layers. The transformer encoder is used to encode the spectral sequence attributes and can be formulated as follows:

$$\begin{aligned} \boldsymbol{F_l'} &= \text{MHA}(\text{LN}(\boldsymbol{F_{l-1}})) + \boldsymbol{F_{l-1}} \\ \boldsymbol{F_l} &= \text{MLP}\left(\text{LN}\left(\boldsymbol{F_l'}\right)\right) + \boldsymbol{F_l'}. \end{aligned} \tag{7}$$

The spectral feature embedding token $\boldsymbol{F_l}$ is the output of the transformer encoder, where l represents the

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encoder depths and $l \in 1, 2, \ldots, k$. Specifically, $\boldsymbol{F_l} = [\boldsymbol{x}'_{\text{hsef_cls}}, \overrightarrow{\boldsymbol{x_1}}', \ldots, \overrightarrow{\boldsymbol{x_i}}', \ldots, \overrightarrow{\boldsymbol{x_c}}']$, where $\boldsymbol{x}'_{\text{hsef_cls}}$ is the generated new class token, which contains the sequence attributes information that reflects the diagnostic spectral characteristics of land covers, and $\overrightarrow{\boldsymbol{x_i}}'$ is the processed ith spectral feature vector. k denotes the number of spectral transformer encoders used in the model. The operations of MHA and MLP have been applied l times alternatively.

C. HS and LiDAR Spatial Transformer

The HS and LiDAR data are heterogeneous, both having been captured over the same area with identical spatial resolution. While HS images provide the topological correlation between land covers, LiDAR images offer rich elevation information. To address the differences between these data sources, we propose two modules, namely, the HSaF and LSaF, to extract spatial features that correspond to the characteristics of the heterogeneous data. The HSaF and LSaF have similar structures, except that the LSaF uses a 1×1 convolution to expand the channel number, as same as HS. This section presents the HSaF module to explain our algorithm in detail. The input HS data are also provided as patch cubes $X^{\dagger} \in R^{c \times h \times w}$, where these patches can be unfolded along the spatial dimensions and rearranged to obtain $\dot{\boldsymbol{X}} \in R^{d \times c}$, where $d = h \times w$ represents the number of pixels in an image patch (i.e., $d = h \times w$). $\dot{\boldsymbol{X}} \in R^{d \times c}$ can be denoted as

$$\dot{\boldsymbol{X}} = \left[\widehat{\boldsymbol{x_1}}, \ldots, \widehat{\boldsymbol{x_i}}, \ldots, \widehat{\boldsymbol{x_d}}\right] \tag{8}$$

where $\widehat{\boldsymbol{x_i}} \in R^{c \times 1}$ denotes the unfold patch for the ith pixel. This way significantly retains spectral sequence information and enables the exploration of relationships between all pixels in each image patch in a global manner. It obtains the correlations between any pixels in the entire image patch, effectively addressing the limitations of CNNs' receptive fields. Subsequently, we also pretend a class token $\boldsymbol{C_2} = \boldsymbol{x}_{\text{hsaf_cls}}$ at the beginning of the sequence for HSaF to aggregate information from all embedding tokens in image patches, which can be formulated by

$$\boldsymbol{G_1} = \text{Concat}\left(\boldsymbol{C_2}, \dot{\boldsymbol{X}}\right) = \left[\boldsymbol{x}_{\text{hsaf_cls}}, \widehat{\boldsymbol{x_1}}, \ldots, \widehat{\boldsymbol{x_i}}, \ldots, \widehat{\boldsymbol{x_d}}\right]. \tag{9}$$

To better capture the contextual information of feature embeddings in a sequence, the order of the tokens needs to be preserved and identified, similar to the HSeF. Thus, it is indispensable to indicate the position of each token by incorporating a positional embedding $\boldsymbol{P_2}$ into the token embeddings $\boldsymbol{G_1}$, which can be expressed as follows:

$$\boldsymbol{G_2} = \boldsymbol{G_1} + \boldsymbol{P_2} \tag{10}$$

where $\boldsymbol{P_2} = [\boldsymbol{p_1}, \ldots, \boldsymbol{p_{d+1}}]$ is regarded as a $c \times (d+1)$ random matrix. Subsequently, the feature embedding $\boldsymbol{G_2}$ is then processed by a transformer encoder consisting of alternating layers of MHA and MLP to capture the relationships between all pixels in the image patch, which can be formulated as follows:

$$\begin{aligned} \boldsymbol{G}'_{\boldsymbol{l}} &= \text{MHA}(\text{LN}(\boldsymbol{G_{l-1}})) + \boldsymbol{G_{l-1}} \\ \boldsymbol{G_l} &= \text{MLP}\left(\text{LN}\left(\boldsymbol{G}'_{\boldsymbol{l}}\right)\right) + \boldsymbol{G}'_{\boldsymbol{l}} \end{aligned} \tag{11}$$

where $l \in \{1, 2, \ldots, k\}$, in which k is encoder layers the same as the HSeF module. $\boldsymbol{G_l}$ is the output feature embedding, and $\boldsymbol{G_l} = [\boldsymbol{x}'_{\text{hsaf_cls}}, \widehat{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widehat{\boldsymbol{x}'_{\boldsymbol{i}}}, \ldots, \widehat{\boldsymbol{x}'_{\boldsymbol{d}}}]$. The newly generated class token $\hat{\boldsymbol{C}}_{\boldsymbol{2}} = \boldsymbol{x}'_{\text{hsaf_cls}}$ can be seen as a pseudo-category of the patch, which obtains the aggregation of in-depth features from a space perspective.

Since the structure of LSaF is similar to HSaF, it would not be repeated here. The output feature embedding of LSaF can be denoted $\boldsymbol{H_l}$ and the class token for LSaF is $\hat{\boldsymbol{C}}_{\boldsymbol{3}} = \boldsymbol{x}'_{\text{lsaf_cls}}$. The generated class token aggregates the elevation information from all patches in LiDAR images and provides complementary information for HS data, making significant contributions to joint classification.

D. HS and LiDAR Collaboration Transformer

The previously mentioned modules (i.e., HSeF, HSaF, and LSaF) are designed to capture specific features unique to each modality, such as spatial and spectral features for HS and elevation information for LiDAR. These modality-specific features indeed provide valuable information for classification performances. However, HS and LiDAR also have shared features, particularly on spatial scales. As both modalities are captured in the same area with the exact spatial resolution, exploring the common features between HS and LiDAR is reasonable.

To achieve this, we use the output feature embeddings from HSaF and LSeF, denoted as $\boldsymbol{G}'_{\boldsymbol{l}} = [\widehat{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widehat{\boldsymbol{x}'_{\boldsymbol{d}}}]$ and $\boldsymbol{H}'_{\boldsymbol{l}} = [\widetilde{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widetilde{\boldsymbol{x}'_{\boldsymbol{d}}}]$, which have removed the class token. We concatenate HS and LiDAR feature embeddings to form an entity $\boldsymbol{Z_1} = \text{Concat}(\boldsymbol{G}'_{\boldsymbol{l}}, \boldsymbol{H}'_{\boldsymbol{l}}) = [\widehat{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widehat{\boldsymbol{x}'_{\boldsymbol{d}}}, \widetilde{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widetilde{\boldsymbol{x}'_{\boldsymbol{d}}}]$ and feed them into a newly devised transformer encoder, named HLCF. We introduce a new class token $\boldsymbol{C_4} = \boldsymbol{x}_{\text{hlcf_cls}}$ to aggregate information from HS and LiDAR features embeddings, which can be formulated by

$$\boldsymbol{Z_1} = \text{Concat}\left(\boldsymbol{C_4}, \hat{\boldsymbol{X}}\right) = \left[\boldsymbol{x}_{\text{hlcf_cls}}, \widehat{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \widehat{\boldsymbol{x}'_{\boldsymbol{d}}}, \widetilde{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots \widetilde{\boldsymbol{x}'_{\boldsymbol{d}}}\right]. \tag{12}$$

To better record the order of HS and LiDAR token embeddings, it is necessary to employ the positional embedding as well. $\boldsymbol{P_4}$ is combined with $\boldsymbol{Z_1}$ by

$$\boldsymbol{Z_2} = \boldsymbol{Z_1} + \boldsymbol{P_4} \tag{13}$$

where $\boldsymbol{P_4} = [\boldsymbol{p_1}, \ldots, \boldsymbol{p_{2d+1}}]$ is regarded as a $c \times (2d+1)$ random matrix. Subsequently, the generated feature embedding $\boldsymbol{Z_2}$ is fed into the transformer encoder composed of alternating layers of MHA and MLP to encoder HS and LiDAR shared features, which can be formulated as follows:

$$\begin{aligned} \boldsymbol{Z}'_{\boldsymbol{l}} &= \text{MHA}(\text{LN}(\boldsymbol{Z_{l-1}})) + \boldsymbol{Z_{l-1}} \\ \boldsymbol{Z_l} &= \text{MLP}\left(\text{LN}\left(\boldsymbol{Z}'_{\boldsymbol{l}}\right)\right) + \boldsymbol{Z}'_{\boldsymbol{l}} \end{aligned} \tag{14}$$

where $\boldsymbol{Z_l}$ is the output feature embedding and $\boldsymbol{Z_l} = [\boldsymbol{x}'_{\text{hlcf_cls}}, \hat{\boldsymbol{x}'_{\boldsymbol{1}}}, \ldots, \hat{\boldsymbol{x}'_{\boldsymbol{d}}}, \ldots, \hat{\boldsymbol{x}'_{\boldsymbol{2d}}}]$. Similar to before, we take out the high semantic class token $\boldsymbol{x}'_{\text{hlcf_cls}}$ separately for the subsequent tokens fusion.

The high-order semantic information and complementary discriminative information in HS and LiDAR feature embeddings have been acquired by the output class tokens

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(i.e., $\boldsymbol{x}'_{\text{hsef_cls}}$, $\boldsymbol{x}'_{\text{hsaf_cls}}$, $\boldsymbol{x}'_{\text{lsaf_cls}}$, and $\boldsymbol{x}'_{\text{hlcf_cls}}$). As aforementioned, HSaF, LSaF, and HLCF concentrate on capturing the HS-modality specific spatial features, LiDAR-modality distinctive elevation information, and HS-LiDAR intermodality shared characteristics. Their corresponding class tokens do not exhibit a significant semantic gap from the spatial perspective. As a result, we integrated three class tokens of HSaF, LSaF, and HLCF by elementwise addition, which can be represented as

$$\boldsymbol{x}_{\text{cls_1}} = \boldsymbol{x}'_{\text{hsaf_cls}} + \boldsymbol{x}'_{\text{lsaf_cls}} + \boldsymbol{x}'_{\text{hlcf_cls}} \tag{15}$$

After contemplating the potential semantic discrepancy between the attributes of spectral sequences and spatial characteristics, we begin by mapping $\boldsymbol{x}'_{\text{hsef_cls}}$ and concatenating it with $\boldsymbol{x}_{\text{cls_1}}$. The resulting vector is then passed through MLP, ultimately producing the final classification results

$$\boldsymbol{x}_{\text{cls_2}} = \text{Concat}\left(\boldsymbol{x}'_{\text{hsef_cls}}, \boldsymbol{x}_{\text{cls_1}}\right) \tag{16}$$

$$\boldsymbol{F}_{\text{out}} = \text{ReLU}\left(\text{MLP}\left(\boldsymbol{x}_{\text{cls_2}}\right)\right) \tag{17}$$

where Linear$(\cdot)$ represents the fully connected layers, rectified linear unit (ReLU) is the activation function, and $\boldsymbol{F}_{\text{out}}$ is the final output vectors and corresponds to the number of classes of specific datasets.

IV. Experiments

In this section, we introduce three widely used HS-LiDAR datasets and provide detailed information on the implementation process. In addition, we conduct extensive experiments, including ablation and parameter sensitivity studies, to evaluate the proposed model's performances quantitatively and qualitatively.

A. Dataset Description

The Houston dataset [60], comprising HS and LiDAR-based digital surface models, was provided by an NSF-funded center at the University of Houston in June 2012. The dataset covered an area of the University of Houston campus and its surrounding regions with a spatial resolution of 2.5 m and consisted of 349 × 1905 pixels. The HS data, captured on June 23, 2012, consisted of 144 spectral bands ranging in the wavelength from 0.38 to 1.05 μm, with 15 classes. The average sensor height above the ground was 5500 ft. The LiDAR data were acquired on June 22, 2012, with a sensor height of 2000 ft. The training and testing samples for each class are listed in Table II. Fig. 3 presents the visualization of the Houston dataset, comprising a false-color image for HS, a grayscale image for LiDAR, and the distribution of training and testing samples on the map.

The AISA Eagle sensor captured the Trento dataset over a rural area in the south of the city of Trento. It consists of 63 spectral bands ranging in the wavelength from 0.43 to 0.99 μm, with a spatial resolution of 1 m and a spatial image size of 600 × 166 pixels. The dataset includes six classes. The LiDAR data were captured using the Optech ALTM 3100EA sensor. The training and testing samples for each class are listed in Table III. Fig. 4 shows the visualization of the Trento dataset, including a false-color image for HS, a grayscale image for LiDAR, and the distribution of training and testing samples on the map.

TABLE II
Number of Training and Testing Samples for the Houston Dataset

Class	Class Name	Training	Testing
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residual	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
	Total	2832	12197

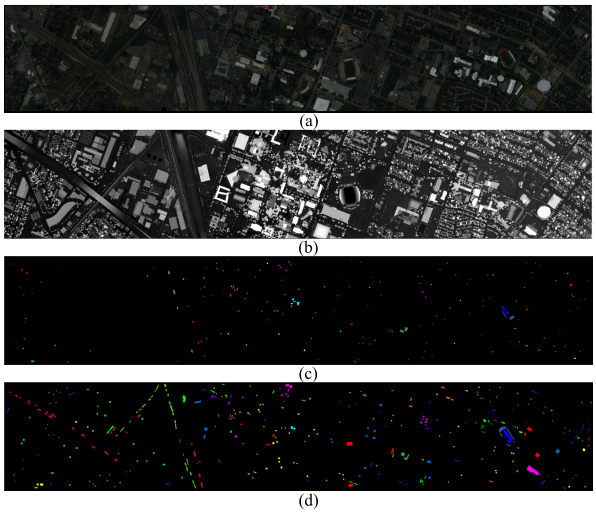


Fig. 3. Houston dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR. (c) Training samples distribution. (d) Testing samples distribution.

The MUUFL Gulfport dataset[1] was collected in November 2010 using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the University of Southern Mississippi Gulf Park campus. The dataset consists of 72 spectral bands with a spatial size of 325 × 220 pixels. In addition, the LiDAR modality consists of two raster data that provide elevation information. The training and testing samples for each class are listed in Table IV. Fig. 5 shows the false-color image for HS, the grayscale image for LiDAR, and the distribution of training and testing samples on the map.

B. Experimental Setup

1) Evaluation Metrics: In this study, we evaluate the quantitative classification performance of HS and LiDAR datasets using three indices: overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). In addition, we visualize

[1]https://github.com/GatorSense/MUUFLGulfport

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TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE TRENTO DATASET

Class	Class Name	Training	Testing
1	Apple	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
	Total	819	29395

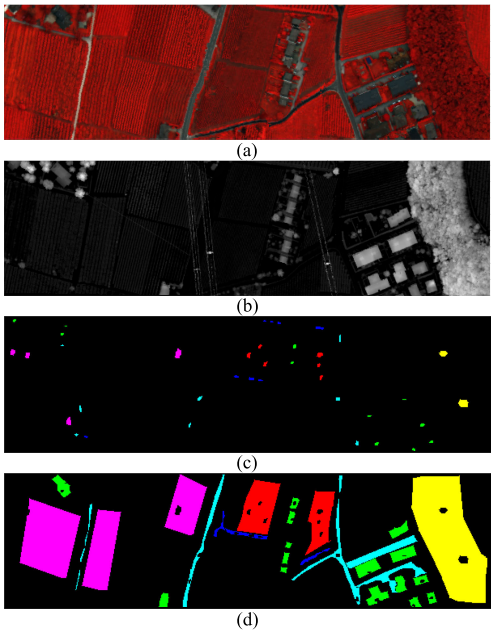


Fig. 4. Trento dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR. (c) Training samples distribution. (d) Testing samples distribution.

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATASET

Class	Class Name	Training	Testing
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building Shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
	Total	1650	52037

all classification maps generated by the different models to make a clear qualitative comparison.

2) Comparison With Other Networks: We selected several conventional methods, i.e., SVM, RF, MLR, and KNN; state-of-the-art CNN-based networks, i.e., CCRNet, S^{2}ENet, and FusAtNet; and ViT-based models, i.e., ViT and SpectralFormer for the comparison experiments. The detailed configurations of these models are listed as follows.

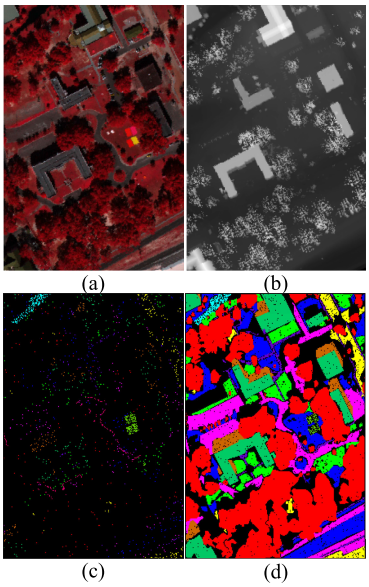


Fig. 5. MUUFL dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR. (c) Training samples distribution. (d) Testing samples distribution.

1) For the SVM, the sklearn-svc toolbox[2] is selected for the implementation of classification tasks. The SVM employs the radial basis function kernel, and its hyperparameters include the regularization parameter and tolerance for stopping criterion, which is empirically set to 100 and $1e^{-7}$, respectively.
2) For the RF, it was implemented using the sklearn-ensemble 1.2.2 toolbox. Four hyperparameters were specified: the number of decision trees, maximum tree depth, minimum samples required at a leaf node, and maximum features at each split. We set the number of decision trees and maximum tree depth to 200 and 10, respectively, while the minimum samples required at a leaf node and maximum features at each split were set to 3 and 10, respectively.
3) For the MLR, it was implemented using the sklearn 1.2.2-linear_model toolbox. For the parameter settings, we chose an L2 penalty and "lbfgs" as the solver for the optimization problem. We set the maximum number of iterations to 5000.
4) For the KNN, the number of nearest neighbors (K) is a crucial factor, as it can significantly impact classification performance. To determine the optimal K value, we performed fivefold cross validation on the training set using the range [5, 10, 15, 20] and selected the best result.
5) The CCRNet[3] architecture comprises three convolutional blocks of 2-D CNN, each consisting of a 2-D convolutional layer, a batch normalization (BN) layer, and a ReLU activation function [30]. The CCR module is also implemented based on the encoder–decoder architecture to enable cross-channel prediction, thereby enabling

[2]https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC
[3]https://github.com/danfenghong/IEEE_TGRS_CCR-Net

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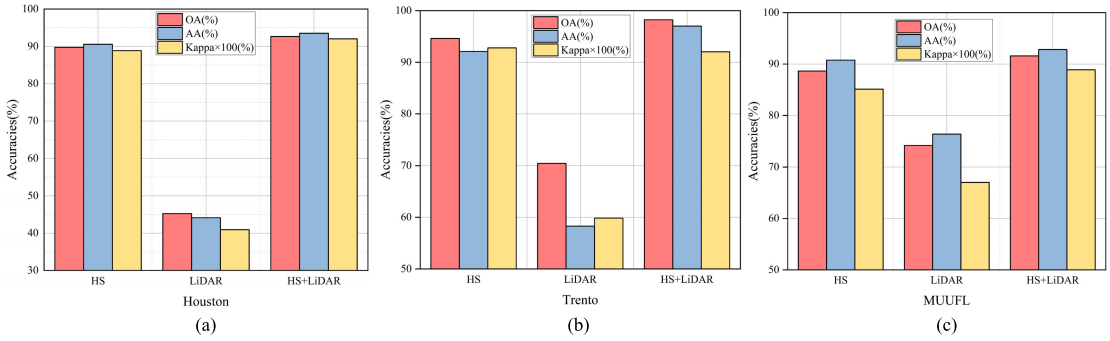


Fig. 6. Ablation studies results in terms of OA (%), AA (%), and Kappa × 100 (%) about the data source usage (i.e., only HS, only LiDAR, and both HS and LiDAR). (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

more comprehensive feature extraction from different modalities.

6) The S^{2}ENet[4] architecture includes two feature enhancement modules: the spatial attention enhancement module and the spectral enhancement module, which enhance the spatial and spectral features of HS and LiDAR, respectively [62]. We followed the remaining hyperparameters consistent with those specified in the original paper.
7) The FusAtNet[5] [41] first utilizes HS modality to generate the attention map using the " SA" mechanism. Then, a cross-attention approach is simultaneously used to harness the LiDAR-derived attention map to accentuate the spatial features. Other hyperparameter configurations are consistent with the original paper.
8) The ViT [56] requires the concatenation of LiDAR and HS data along the channel dimension. The network structure is followed by including only the encoder layers in the joint classification of HS and LiDAR data.
9) For the SpectralFormer[6] [43], we concatenate the LiDAR data to the HS data along the channel dimension, and the backbone architecture is identical to that of ViT to ensure a fair comparison. The embedded spectrum, with 64 units, is fed into five cascaded transformer encoder blocks, each containing four head SA layers, an MLP layer with eight hidden dimensions, and a GeLU nonlinear activation function.

3) Implementation Details: The experiments were conducted on a computer equipped with a Ryzen 3600X processor, 64-GB RAM, and an RTX3080Ti-12GB GPU. Each experiment was performed independently five times, and the mean value was selected as the final result. We implemented the proposed model using Pytorch and utilized the Adam optimization to optimize the parameters with a batch size of 64. The learning rate was initialized at $1e^{-4}$ and decayed by multiplying a factor of 0.1 after each 1/50th of the total epochs. The total number of training epochs was set to 900. To prevent the overfitting phenomenon, we employed the early stopping strategy if the model parameters had not been updated for more than 50 epochs.

[4]https://github.com/likyoo/Multimodal-Remote-Sensing-Toolkit
[5]https://github.com/ShivamP1993/FusAtNet-Dual-Attention-based-SpectroSpatial-Multimodal-Fusion-Network-for-Hyperspectral-and-LiDAR-
[6]https://github.com/danfenghong/IEEE_TGRS_SpectralFormer

C. Model Analysis

1) Ablation Studies: We conducted ablation studies to assess the contributions of the two heterogeneous data, HS and LiDAR, separately toward classification. The results are shown in Fig. 6, and it can be seen that HS data significantly enhance classification performances compared to LiDAR data, which is because HS images contain a large number of spectral bands that offer rich spatial and spectral characteristics of land covers, while LiDAR provides limited elevation information, which is less favorable for accurate classification. Nonetheless, integrating both data (i.e., HS + LiDAR) achieved better performances than using a single data source. The results demonstrate that HS and LiDAR complement each other and can enhance classification results to some extent.

We investigated the performance of our proposed model by adding different transformer structures (HSaF, LSaF, HLCF, and HSeF) to the networks in a stepwise manner. Table V summarizes the results of all combinations of the four transformer structures on three datasets. A single structure alone often led to unsatisfactory classification results due to inadequate feature mining. However, combining two structures resulted in better performance, such as the HSeF and HSaF combination, which significantly improved performance (around 1%–2% OA). In addition, the joint use of HSaF and HLCF led to a dramatic increase in performance (around 4%–6% OA). Surprisingly, combining three structures did not result in the expected performance improvement, possibly due to overfitting caused by inappropriate combinations. Furthermore, transformer structures with spatial features outperformed those with spectral features (around 2%–5% OA). Finally, by integrating all four transformer structures, we captured the common and specific characteristics of HS and LiDAR and achieved the best performance in terms of OA, AA, and Kappa across all three datasets.

2) Parameter Sensitivity Analysis: In the training process, the number of encoder depths and attention heads of transformer structures are crucial parameters that affect the joint classification of HS and LiDAR. Therefore, exploring the appropriate parameter ranges is necessary to achieve promising classification results. Similarly, we also conduct an ablation study on three datasets regarding OA. To ensure consistency, the number of depths and heads for all transformer structures are fixed, ranging from 2 to 7, with an interval of 1. Fig. 7 shows the OA with variations in the number of encoder depths

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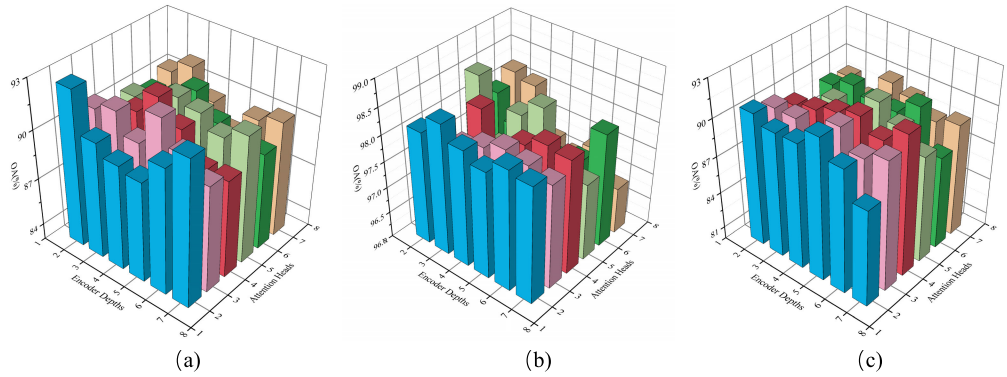


Fig. 7. Impact of a different number of encoder depths and attention heads in terms of OA (%) on three datasets. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

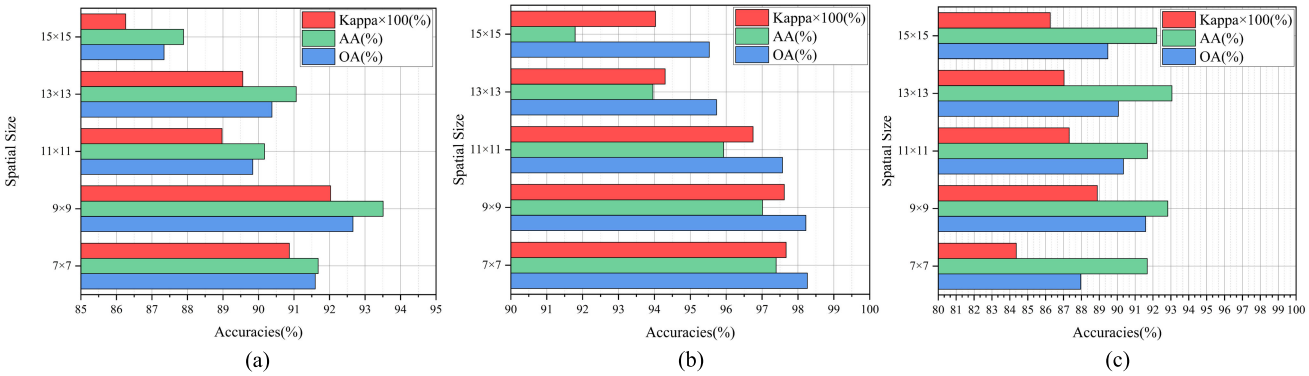


Fig. 8. Impact of different spatial sizes of image cubes in terms of OA (%), AA (%), and Kappa × 100 (%) on three datasets. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

TABLE V
ABLATION STUDIES RESULT ON DIFFERENT COMBINATIONS OF MULTIPLE TRANSFORMERS FOR THREE DATASETS

Moudules	Metrics	Different Combinations														
HSeF		✓	-	-	-	✓	✓	✓	-	-	-	✓	✓	✓	-	✓
HSaF		-	✓	-	-	✓	-	-	✓	✓	-	✓	✓	-	✓	✓
LSaF		-	-	✓	-	-	✓	-	✓	-	✓	✓	-	✓	✓	✓
HLCF		-	-	-	✓	-	-	✓	-	✓	✓	-	✓	✓	✓	✓
Houston	OA(%)	86.33	88.36	45.24	91.51	89.74	90.91	89.87	91.34	91.46	91.54	89.05	90.83	91.65	91.37	92.66
	AA(%)	86.74	89.14	44.13	91.23	90.55	90.82	90.81	90.76	91.99	91.7	89.14	91.55	92.48	91.71	93.51
	Kappa×100(%)	85.16	87.37	40.92	90.78	88.86	90.13	89.01	89.56	90.72	90.81	88.12	90.05	90.93	90.62	92.03
Trento	OA(%)	92.02	97.19	70.44	97.87	94.62	97.48	97.55	98.04	98.32	97.93	97.98	98.27	97.66	97.86	98.23
	AA(%)	87.68	92.63	58.28	96.67	92.10	95.20	94.81	96.83	96.82	97.21	96.88	97.57	95.79	96.90	97.01
	Kappa×100(%)	89.74	93.56	59.84	97.15	92.79	96.62	96.72	97.52	97.75	97.23	97.29	97.68	96.87	97.14	97.62
MUUFL	OA(%)	83.24	87.90	74.16	90.58	88.65	85.24	86.69	89.51	90.01	90.26	86.97	89.09	88.32	90.45	91.59
	AA(%)	85.61	90.81	76.37	92.39	90.77	87.7	89.66	91.47	91.61	92.54	88.82	90.84	89.34	91.61	92.83
	Kappa×100(%)	78.24	84.23	67.01	87.64	85.13	80.90	82.72	86.27	86.89	87.23	83.02	85.74	84.70	87.44	88.88

and attention heads. The best results for the Houston dataset are obtained when the encoder depths and attention heads are set to 2, which seems counterintuitive. In general, deeper features can be captured with increased encoder layers and attention heads. However, increasing the number of layers and heads can also lead to overfitting, and the OA may fluctuate irregularly. The optimal number of the encoder and attention heads for the Trento dataset is 3 and 2, respectively. Similarly, no definite trend in classification performances is observed with respect to the change in parameters. For the MUUFL dataset, the best results are achieved when the number of the encoder and attention heads is set to 5 and 2, respectively.

Furthermore, the spatial size is a crucial parameter that can significantly influence the performance of the proposed model. To evaluate the impact of this parameter, we tested multiple spatial sizes from a candidate set 7×7, 9×9, 11×11, 13×13, and 15×15. Fig. 8 shows the OA, AA, and Kappa on the three datasets with different spatial sizes. The results show that the classification accuracies on the MUUFL dataset increase and remain relatively stable as

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TABLE VI
COMPUTATIONAL COMPLEXITY RESULTS IN TERMS OF NETWORK PARAMETERS (M), FLOPS (G), TRAINING TIME (S), AND TESTING TIME (S) FOR SEVERAL DL-BASED MODELS ON HOUSTON, TRENTO, AND MUUFL DATASETS

Dataset	Complexity	S^2ENet	FusAtNet	CCRNet	ViT	SpectralFormer	Proposed
Houston	Parameters(M)	0.271	36.904	0.071	0.089	0.224	0.34
	Flops(G)	0.848	221.612	0.399	0.808	1.413	2.32
	Training Time(s)	258.03	1329.44	327.9	309.49	399.01	253.30
	Testing Time(s)	1.26	5.15	1.53	1.76	2.01	1.50
Trento	Parameters(M)	0.177	36.242	0.059	0.089	0.122	0.15
	Flops(G)	0.555	216.552	0.021	0.357	0.496	1.01
	Training Time(s)	75.18	394.22	78.20	76.2	95.81	77.25
	Testing Time(s)	2.87	12.44	4.14	4.85	3.74	5.02
MUUFL	Parameters(M)	0.179	36.26	0.062	0.089	0.123	0.15
	Flops(G)	0.559	216.65	0.042	0.362	0.505	1.03
	Training Time(s)	138.89	773.52	189.3	155.44	175.74	151.8
	Testing Time(s)	4.84	3.60	7.56	4.94	6.31	6.74

the spatial sizes change from 7×7 to 15×15. However, the OA of the Houston and Trento datasets is greatly affected by this parameter. Specifically, the OA increases initially and dramatically decreases from a spatial size of 7×7 to 9×9 on the Houston dataset. The OA then tends to fluctuate up and down with the increase in spatial sizes. On the Trento dataset, the OA decreases continuously from a spatial size of 7×7 to 15×15. Based on the experimental results, the reasonable spatial sizes for the Houston, Trento, and MUUFL datasets are 9×9, 7×7, and 9×9, respectively.

3) Computational Complexity: We performed extensive experiments to evaluate the computational complexity of CNN- and transformer-based models. Specifically, we evaluated these models based on four metrics: network parameters, floating points (Flops), training time, and testing time. We used the parameter configuration of the compared models from the original papers to obtain the best results. The detailed experimental results are reported in Table VI. Among the CNN-based models (e.g., S^2ENet and CCRNet), there were no significant differences in terms of computational complexity, except for FusAtNet, which has multiple convolutional layers and a large number of convolutional filters. The ViT-based models, due to their massive number of parameters and complex MHA calculation operations, performed worse than the CNN-based models in terms of computational complexity. Our proposed model also has many parameters and a relatively high time complexity. However, it outperformed the two other transformer-based models, i.e., SpectralFormer and ViT. Moreover, our model's training and testing time on the three datasets was lower than those of FusAtNet and CCRNet. Overall, our model is highly competitive in achieving lightweight and promising classification results.

4) Training Ratio: We randomly selected different proportions of training samples (20%, 50%, 70%, 90%, and 100%) from the training set of Houston, Trento, and MUUFL datasets over five runs. We then evaluated the performance of the compared models, including S^2ENet, FusAtNet, CCRNet, ViT, SpectralFormer, and our proposed MTNet, regarding OA, AA, and Kappa. The average results are reported in Fig. 9. A clear trend can be observed in the OA's results, where the classification performance of all models improves gradually with an increase in the percentage of training samples. These results demonstrate that our proposed model achieves promising performance with limited training samples compared to other state-of-the-art models. Moreover, the proposed MTNet's OA tends to stabilize when more training samples are used, exhibiting the promising robustness of the proposed model.

5) Quantitative Results and Analysis: The quantitative results for the Houston, Trento, and MUUFL datasets in terms of OA, AA, Kappa, and accuracies for each class are presented in Tables VII–IX. In general, conventional classifiers, such as SVM, RF, MLR, and KNN, achieve comparable classification performances on all three datasets, except for MLR's accuracies on MUUFL, which are substantially lower (around 13%–14% OA) than those obtained by SVM and RF. CNN-based networks, including S^2ENet, FusAtNet, and CCRNet, have achieved promising performances due to the powerful feature representations of CNNs and outperform the aforementioned conventional classifiers. Transformer-based models, such as ViT and SpectralFormer, can extract spectral representations from HS data without convolutional or fully connected operations. While these models can capture global sequence information from the spectral perspective, their ability to model key factors from the spatial perspective is limited. Not surprisingly, the results obtained by these transformer-based models are inferior to those of CNN-based models (around 8%–10% OA). For heterogeneous data such as HS and LiDAR data, ViT and SpectralFormer are not proficient in obtaining useful information from LiDAR data to enhance the classification tasks. To address the limitations of these models, we have designed different transformer-based structures carefully to extract common and individual features of HS and LiDAR data better in the proposed MTNet, resulting in better classification performances than those of the CNN-based backbone networks. Moreover, for some challenging classes such as Highway, Parkinglot1, and Parkinglot2, the other compared methods face classification bottlenecks, while our proposed model exhibits robust generalization ability (81.47%, 85.4%, and 88.42% OA on Highway, Parkinglot1, and Parkinglot2 classes). In addition, for some classes on the Houston dataset

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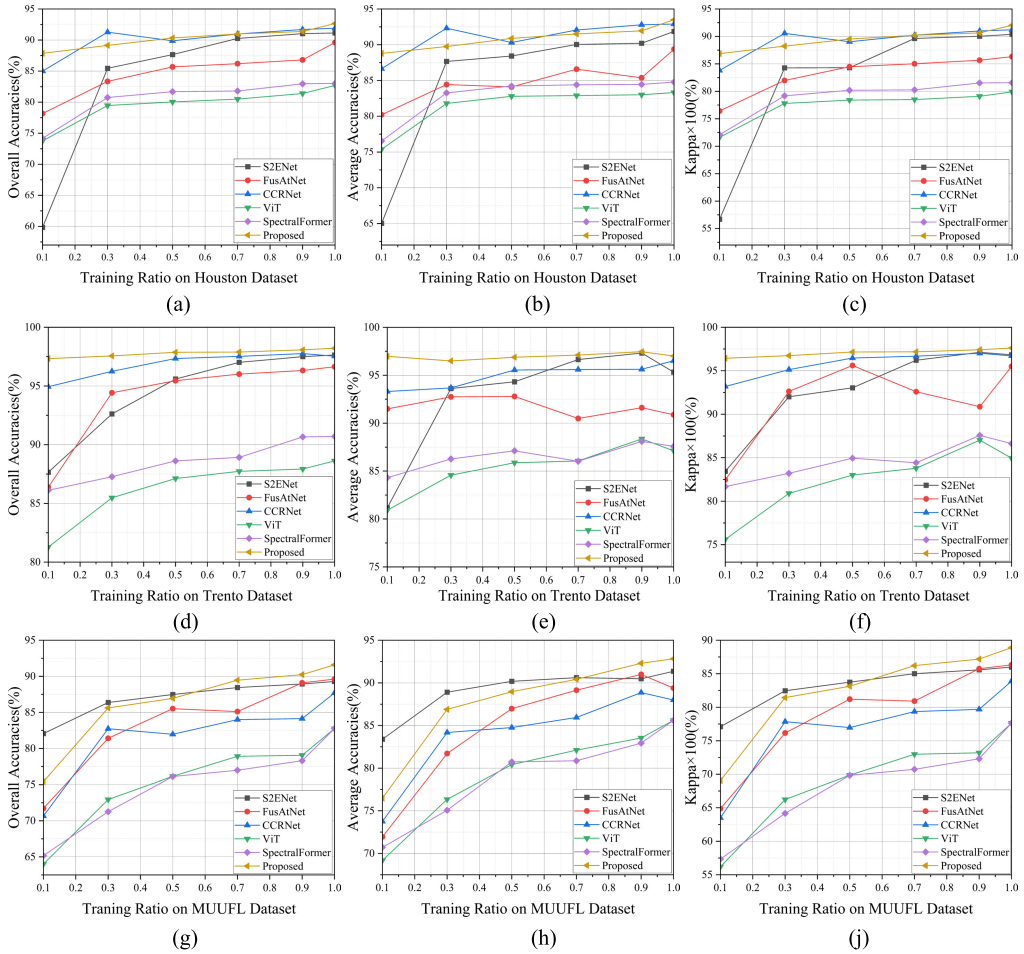


Fig. 9. Impact of training ratio based on the original training samples selection in terms of OA (%), AA (%), and Kappa × (%) for several DL-based models for three datasets. (a)–(c) OA, AA, and Kappa on Houston dataset, respectively. (d)–(f) OA, AA, and Kappa on Trento dataset, respectively. (g)–(i) OA, AA, and Kappa on MUUFL dataset, respectively.

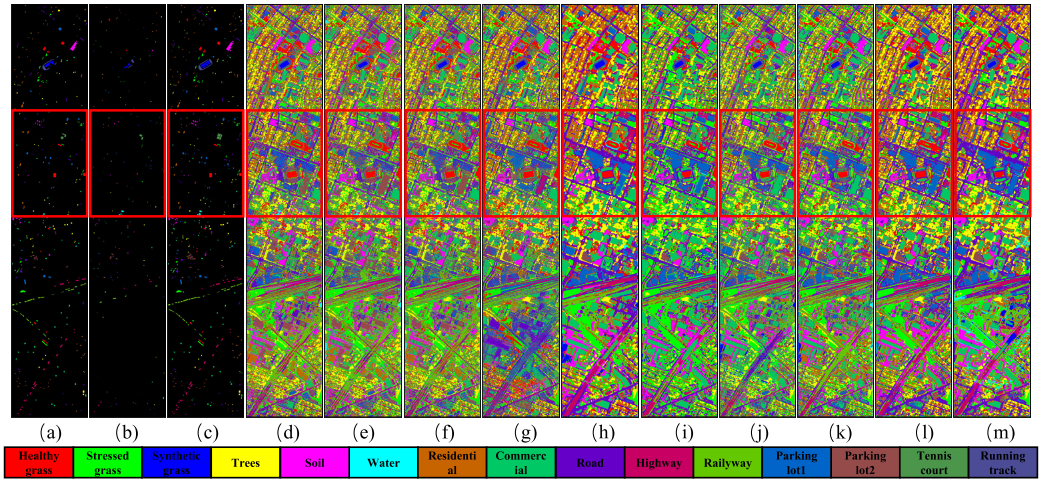


Fig. 10. Classification maps on Houston dataset obtained by different models. (a) Ground truth. (b) Training samples distribution map. (c) Testing samples distribution map. (d) SVM. (e) RF. (f) MLR. (g) KNN. (h) S^2ENet. (i) FusAtNet. (j) CCRNet. (k) ViT. (l) SpectralFormer. (m) Proposed.

(e.g., synthetic grass, trees, soil, tennis court, and running track), the proposed MTNet achieved 100% OA, demonstrating the robust and accurate land cover classification capability.

D. Visual Evalutaion

We also evaluated qualitatively by visualizing the classification maps obtained from different methods. The obtained results for Houston, Trento, and MUUFL datasets are presented in Figs. 10–12. The conventional classifiers (i.e., SVM, RF, MLR, and KNN) tended to generate salt and pepper noise in all three datasets, indicating their inability to identify object materials accurately. In contrast, CNN-based models (i.e., S^2ENet, FusAtNet, and CCRNet) produced relatively smooth classification maps, due to their powerful nonlinear data fitting

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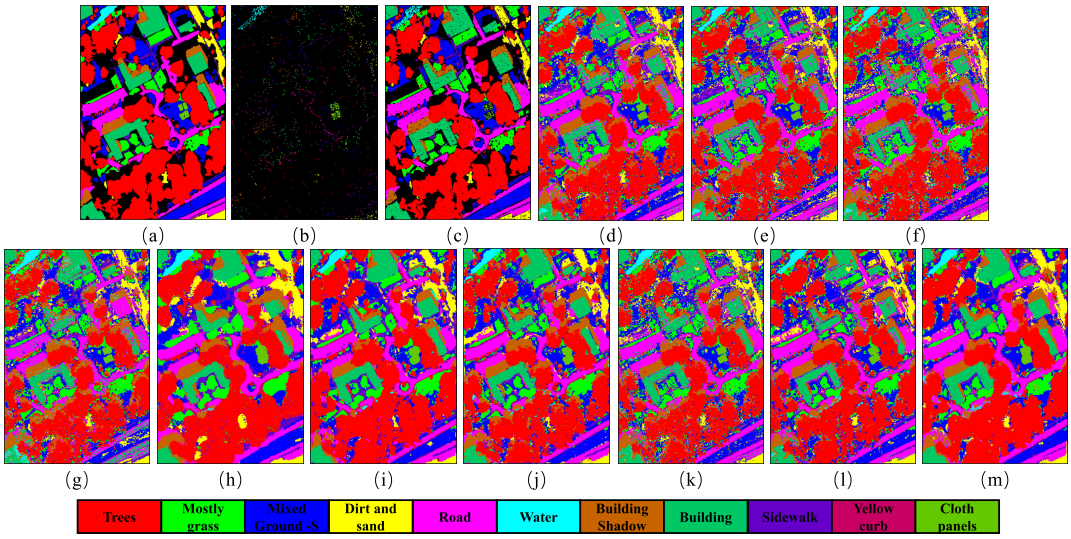


Fig. 11. Classification maps on Trento dataset obtained by different models. (a) Ground truth. (b) Training samples distribution map. (c) Testing samples distribution map. (d) SVM. (e) RF. (f) MLR. (g) KNN. (h) S^2ENet. (i) FusAtNet. (j) CCRNet. (k) ViT. (l) SpectralFormer. (m) Proposed.

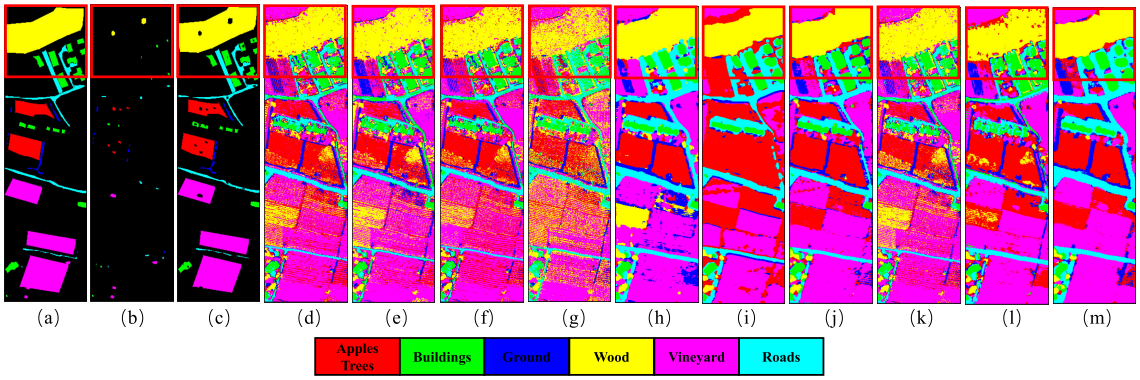


Fig. 12. Classification maps on MUUFL dataset obtained by different models. (a) Ground truth. (b) Training samples distribution map. (c) Testing samples distribution map. (d) SVM. (e) RF. (f) MLR. (g) KNN. (h) S^2ENet. (i) FusAtNet. (j) CCRNet. (k) ViT. (l) SpectralFormer. (m) Proposed.

TABLE VII

CLASSIFICATION RESULTS IN TERMS OF OA (%), AA (%), AND KAPPA × 100 (%) FOR DIFFERENT METHODS ON THE HOUSTON DATASET. THE BEST RESULTS ARE SHOWN IN BOLD. THE SUBOPTIMAL RESULTS ARE SHOWN WITH UNDERLINED TYPE

Class	Conventional Classifiers				Convolutional Neural Networks			Transformer Networks		
No.	SVM	RF	MLR	KNN	S^2ENet	FusAtNet	CCRNet	ViT	SpectralFormer	Proposed
1	86.25	86.01	**95.84**	85.93	83.38	82.43	84.33	82.81	81.58	83.10
2	97.13	98.09	96.41	96.57	95.49	98.4	**99.81**	97.18	96.33	99.44
3	99.71	97.56	99.71	99.57	97.23	87.52	97.62	**100.00**	94.65	**100.00**
4	98.63	97.83	93.09	98.55	99.53	97.35	**100.00**	96.88	97.63	**100.00**
5	98.15	96.46	96.78	96.78	**100.00**	99.34	97.63	97.92	99.81	**100.00**
6	97.54	96.92	99.69	96.92	**100.00**	97.90	96.50	95.10	86.71	95.80
7	89.04	85.09	82.33	84.15	85.45	93.84	**98.60**	82.00	86.10	92.44
8	52.73	47.03	57.64	56.67	93.54	94.3	79.68	53.75	75.88	**96.68**
9	81.31	71.65	69.01	72.84	83.38	87.91	81.68	69.50	70.54	**86.69**
10	69.27	56.32	62.59	61.94	90.73	62.45	**81.56**	61.00	49.13	81.47
11	88.42	82.35	77.33	85.26	**93.83**	82.73	90.32	77.80	82.35	93.26
12	64.64	44.20	69.42	52.55	80.60	**87.22**	85.21	63.30	75.41	85.40
13	69.30	60.55	77.83	40.51	77.19	81.05	88.77	74.74	82.11	**88.42**
14	99.07	98.60	99.77	97.43	99.19	99.60	97.98	**100.00**	**100.00**	**100.00**
15	98.33	96.52	97.27	98.48	98.31	93.02	69.77	97.89	93.45	**100.00**
OA(%)	84.47	79.00	82.67	80.68	91.14	88.94	89.68	80.50	83.01	**92.66**
AA(%)	85.97	81.01	84.98	81.62	91.86	89.67	89.96	83.32	84.78	**93.51**
Kappa×100(%)	83.22	77.35	81.28	79.12	90.39	89.99	88.79	78.90	81.57	**92.03**

ability. The emerging network architecture of transformers, such as ViT and SpectralFormer, also exhibits excellent feature representation capabilities similar to CNNs. However, these structures are particularly suitable for capturing spectral features from HS but tend to ignore the elevation information in LiDAR data. Although the classification performances of

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TABLE VIII

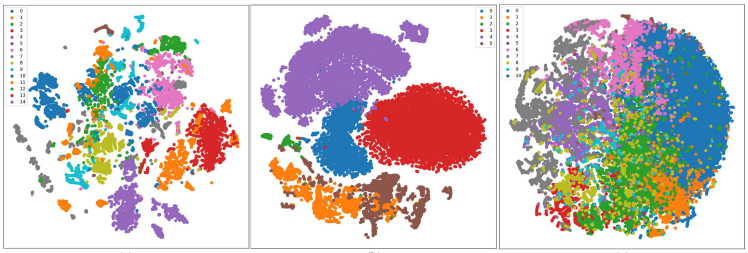
CLASSIFICATION RESULTS IN TERMS OF OA (%), AA (%), AND KAPPA × 100 (%) FOR DIFFERENT METHODS ON THE TRENTO DATASET. THE BEST RESULTS ARE SHOWN IN BOLD. THE SUBOPTIMAL RESULTS ARE SHOWN WITH UNDERLINED TYPE

Class	Conventional Classifiers				Convolutional Neural Networks			Transformer Networks		
No.	SVM	RF	MLR	KNN	S^2ENet	FusAtNet	CCRNet	ViT	SpectralFormer	Proposed
1	93.11	85.40	76.30	93.26	96.16	**100.00**	**100.00**	88.02	91.06	96.70
2	84.98	86.12	65.45	83.53	**100.00**	99.03	99.75	84.23	89.45	99.28
3	**97.70**	87.47	93.95	97.29	92.25	61.50	94.39	96.52	95.19	96.52
4	97.04	95.59	80.43	93.73	99.99	98.55	**100.00**	96.78	93.67	99.91
5	77.63	80.70	62.29	66.71	**99.82**	96.32	96.74	77.58	94.69	99.52
6	70.51	64.37	72.56	69.12	83.78	89.91	88.24	69.17	61.40	**90.14**
OA(%)	85.83	84.74	71.52	80.77	97.64	96.64	97.54	84.82	89.95	**98.23**
AA(%)	86.83	83.27	75.16	83.94	95.33	90.88	96.52	85.38	87.58	**97.01**
Kappa×100(%)	81.43	79.79	62.33	75.05	96.84	95.51	96.71	79.94	86.62	**97.62**

TABLE IX

CLASSIFICATION RESULTS IN TERMS OF OA (%), AA (%), AND KAPPA × 100 (%) FOR DIFFERENT METHODS ON THE MUUFL DATASET. THE BEST RESULTS ARE SHOWN IN BOLD. THE SUBOPTIMAL RESULTS ARE SHOWN WITH UNDERLINED TYPE

Class	Conventional Classifiers				Convolutional Neural Networks			Transformer Networks		
No.	SVM	RF	MLR	KNN	S^2ENet	FusAtNet	CCRNet	ViT	SpectralFormer	Proposed
1	82.59	82.1	80.59	81.97	87.72	93.35	90.06	80.42	85.56	**94.39**
2	81.19	79.41	81.71	83.09	85.7	84.93	66.46	81.94	75.75	**88.06**
3	75.02	68.61	72.25	64.97	80.69	77.12	**82.81**	69.04	74.08	82.07
4	85.82	82.15	84.99	76.07	94.45	92.96	86.87	85.98	86.40	**96.30**
5	87.62	84.63	73.71	88.04	89.28	86.65	90.91	86.83	88.83	**92.31**
6	92.27	93.78	97.42	89.48	**100.00**	98.73	99.05	93.04	98.10	99.37
7	89.12	86.97	85.54	85.40	93.23	89.25	**94.29**	84.11	88.72	92.80
8	78.89	72.79	72.00	71.25	90.74	95.12	**94.52**	83.78	77.44	92.27
9	78.27	68.59	71.99	59.64	77.09	**85.02**	76.03	70.93	74.41	84.37
10	97.27	94.54	95.63	86.34	93.94	81.82	87.88	**100.00**	93.94	**100.00**
11	98.51	98.14	98.88	97.03	99.16	98.32	99.16	**100.00**	98.32	99.16
OA(%)	82.19	79.47	78.18	78.31	87.48	89.61	87.69	80.50	82.75	**91.59**
AA(%)	86.05	82.88	83.16	80.30	90.18	89.39	88.00	85.10	85.60	**92.83**
Kappa×100(%)	77.31	73.89	72.34	73.17	83.74	86.33	83.88	75.02	77.63	**88.88**



(a) (b) (c)

Fig. 13. 2-D graphical visualization of the HS features extracted by the proposed MTNet through t-SNE on three datasets. (a) Houston dataset. (b) Trento dataset. (c) MUUFL dataset.

these transformer-based models have surpassed those of conventional classifiers, there is still a significant gap compared to CNN-based models. We used red rectangular boxes to mark the ROI to highlight the differences in classification maps between different models on Houston and MUUFL datasets, further evaluating the classification performance. Remarkably, our proposed model showed more realistic and finer details in the ROIs, with fewer noisy points than other compared models. In addition, the MTNet avoided oversmoothness in edges or small semantic objects, yielding more accurate classification maps.

In addition, the raw HS high-dimensional input patches were reduced to 2-D using the t-distributed stochastic neighbor embedding (t-SNE) technique, and the visualization results for Houston, Trento, and MUUFL datasets are presented in Fig. 13. It can be seen that samples of the same class were clustered together, and the gaps between samples of different classes were widened, demonstrating the outstanding discriminative ability of the proposed MTNet.

V. CONCLUSION

This article presents an MTNet for HS and LiDAR classification. Rather than relying on CNNs for feature extraction,

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our proposed method utilizes multiple transformers to capture individual and shared features of the two modalities. HS data contain numerous spectral bands and abundant spectral information. To capture spectrally local sequence signatures from neighboring bands of HS images, we developed the HSeF feature extraction module, which learns the targeted feature and yields a group-spectral embedding. To capture spatial and elevation information of HS and LiDAR data, we designed HSaF and LSaF modules, which capture pixelwise feature embedding relationships in a global manner. Finally, we combine the feature embeddings of both modalities to identify shared modality characteristics, enhancing the classification performance. We evaluate the proposed MTNet on three popular HS and LiDAR datasets and demonstrate that it outperforms conventional methods (i.e., SVM, RF, MLR, and KNN), state-of-the-art CNN-based networks (i.e., S^2ENet, FusAtNet, and CCRNet), and transformer-based models (i.e., ViT and SpectralFormer) from both quantitative and visual analyses.

Nonetheless, our proposed method has certain limitations that necessitate further exploration. First, we require a more in-depth examination of weight assignment in multiple transformer modes to improve the classification performance. Second, future studies should simplify the model's parameters while maintaining its promising classification performance. In addition, we intend to reduce reliance on labeled samples and explore semisupervised or unsupervised classification methods.

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